Covering Families for Complete Bipartite Graphs with DP-Coloring

Illia Siutkin

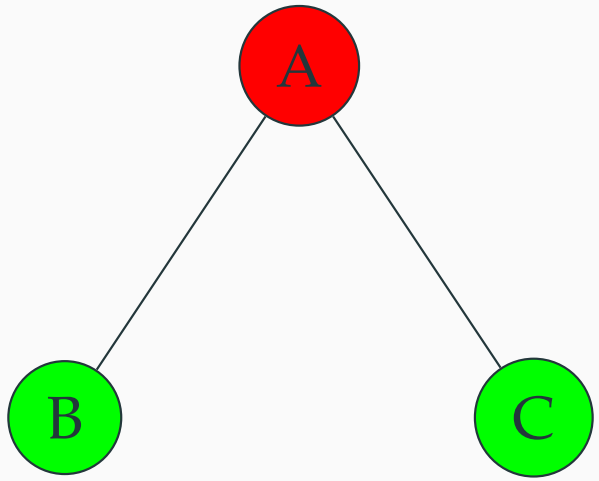
October 19, 2026

Lake Forest College

Introduction to DP-Coloring

Graph Coloring

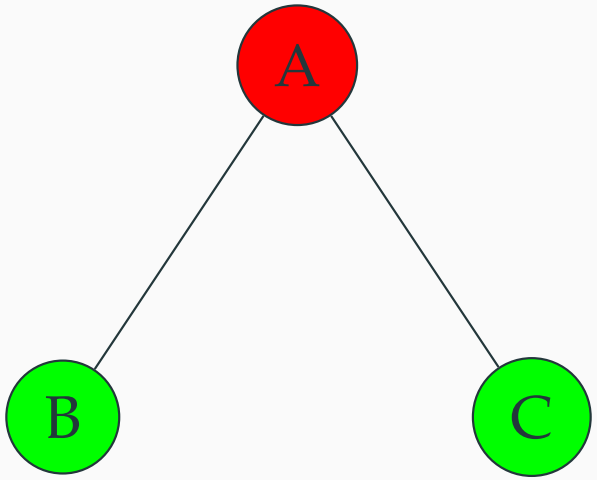
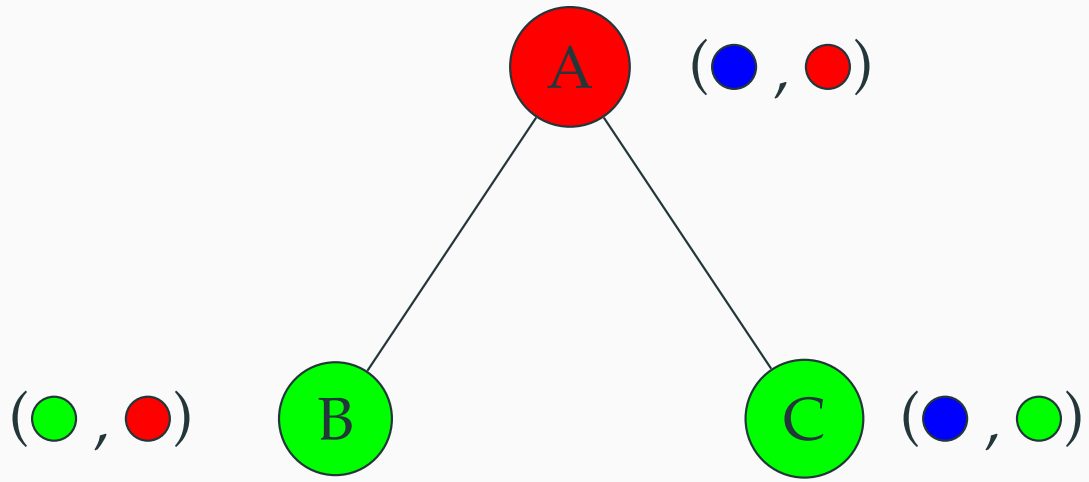
Allocation of colors to the vertices. In a proper coloring, adjacent vertices receive different colors.



Chromatic Number

Definition: $\chi(G)$ = minimum number of colors needed to color the graph properly

Here, $\chi(G) = 2$



List Coloring